

UNIT 5: DISK SCHEDULING & INODE SOLVED NUMERICAL EXAMPLES

Course Pattern: TE IT - 2024 Pattern (SPPU)

Academic Term: Semester V (2026)

Document Type: Solved Numerical Guide

Course Code: PCC-301-ITT

Problem 1: Disk Scheduling Cylinder Head Movements

INTUITION:

Algorithm Rules & Calculations:

- Total Head Movement is calculated by adding the absolute difference $|C_{\text{next}} - C_{\text{current}}|$ for each head movement step.
- **FCFS**: Sequential queue processing.
- **SSTF**: Move to request with minimum $|C_{\text{next}} - C_{\text{current}}|$.
- **SCAN**: Move in initial direction up to boundary cylinder (0 or 199), then reverse.
- **LOOK**: Move in initial direction up to max/min request cylinder, then reverse (does not go to physical disk boundary).

Problem Statement

Consider a disk queue with requests for I/O to blocks on cylinders: 98, 183, 37, 122, 14, 124, 65, 67

The disk head is currently at cylinder **53**, moving toward **higher cylinders** (larger cylinder numbers). Total cylinders on disk: 0 to 199.

Calculate the total number of head movements (in cylinders) using:

- (a) FCFS Disk Scheduling
- (b) SSTF Disk Scheduling
- (c) SCAN Disk Scheduling
- (d) LOOK Disk Scheduling

Step-by-Step Solution

1.2.1 (a) FCFS Algorithm

- Head trajectory: 53to98to183to37to122to14to124to65to67

- Movements:

1. $|98 - 53| = 45$
2. $|183 - 98| = 85$
3. $|37 - 183| = 146$
4. $|122 - 37| = 85$
5. $|14 - 122| = 108$
6. $|124 - 14| = 110$
7. $|65 - 124| = 59$
8. $|67 - 65| = 2$

- **Total Head Movement (FCFS):** $45 + 85 + 146 + 85 + 108 + 110 + 59 + 2 = \mathbf{640}$ cylinders

1.2.2 (b) SSTF Algorithm

- Starting at 53, active queue: {14, 37, 65, 67, 98, 122, 124, 183}
 - From 53, closest is 65: $|65 - 53| = 12$
 - From 65, closest is 67: $|67 - 65| = 2$
 - From 67, closest is 37: $|37 - 67| = 30$
 - From 37, closest is 14: $|14 - 37| = 23$
 - From 14, closest is 98: $|98 - 14| = 84$
 - From 98, closest is 122: $|122 - 98| = 24$
 - From 122, closest is 124: $|124 - 122| = 2$
 - From 124, closest is 183: $|183 - 124| = 59$
- Total Head Movement (SSTF):** $12 + 2 + 30 + 23 + 84 + 24 + 2 + 59 = \text{bold}\{236 \text{ cylinders}\}$

1.2.3 (c) SCAN (Elevator) Algorithm (Moving High to Boundary 199)

- Head moves toward higher cylinders up to boundary 199, then reverses down to remaining requests:
- Path: 53to65to67to98to122to124to183to **bold}\{199\}**to37to14
- Calculation formula: $(199 - 53) + (199 - 14) = 146 + 185$
- Total Head Movement (SCAN):** **bold}\{331 \text{ cylinders}\}**

1.2.4 (d) LOOK Algorithm (Moving High to Max Request 183)

- Head moves toward higher cylinders up to max request 183 (does NOT go to boundary 199), then reverses down to 14:
- Path: 53to65to67to98to122to124to **bold}\{183\}**to37to14
- Calculation formula: $(183 - 53) + (183 - 14) = 130 + 169$
- Total Head Movement (LOOK):** **bold}\{299 \text{ cylinders}\}**

Problem 2: Unix Inode Maximum File Size Math**Problem Statement**

Consider a Unix file system using an Inode with:

- 12 Direct block pointers
- 1 Single indirect pointer
- 1 Double indirect pointer
- 1 Triple indirect pointer

Disk block size = 1 KB = 1024 bytes. Each block pointer requires 4 bytes.

Calculate the maximum file size supported by this Inode architecture.

Step-by-Step Solution**2.2.1 1. Block Pointer Capacity**

- Number of pointers per indirect block = $\frac{1024}{4} = 256$ pointers.

2.2.2 2. Capacity Contribution by Pointer Type

- Direct Blocks (12 pointers):**
 - $12 * 1 \text{ KB} = 12 \text{ KB}$
- Single Indirect Block (1 pointer):**
 - $256 * 1 \text{ KB} = 256 \text{ KB}$
- Double Indirect Block (1 pointer):**
 - $256 * 256 * 1 \text{ KB} = 65,536 \text{ KB} = 64 \text{ MB}$
- Triple Indirect Block (1 pointer):**
 - $256 * 256 * 256 * 1 \text{ KB} = 16,777,216 \text{ KB} = 16 \text{ GB}$

2.2.3 3. Maximum Supported File Size

- Max File Size = 12 KB + 256 KB + 64 MB + 16 GB $\approx$ **16.064 GB**
-

End of Operating Systems Unit 5 Solved Examples — SPPU TE IT 2024 Pattern